NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
Marshall Space Flight Center
JUSTIFICATION FOR OTHER THAN FULL AND OPEN COMPETITION (JOFOC)
For Determinant Assembly for the Space Launch System Block 1B Payload Adapter

1. **Federal Acquisition Regulation (FAR) 6.303-2(b)(1) – Identification of the agency and the contracting activity, and specific identification of the document as a "Justification for other than full and open competition."**

   This document is a justification for other than full and open competition prepared by National Aeronautics and Space Administration (NASA) George C. Marshall Space Flight Center (MSFC).

   The procuring agency is the National Aeronautics and Space Administration (NASA) and the contracting activity is MSFC located in Huntsville, AL.

2. **FAR 6.303-2(b)(2) – The nature and/or description of the action being approved:**

   This justification provides the rationale for contracting by other than full and open competition to award a sole source contract to Shape Fidelity, Inc.

   This contract action will provide for assembly of a Payload Adapter (PLA) in support of the Space Launch System (SLS) rocket.

3. **FAR 6.303-2(b)(3) – A description of the supplies or services required, to meet the Agency's needs (including the estimated value):**

   The services being procured involve the assembly of SLS B1B PLA on site at NASA/MSFC. This requirement is for similar services that were procured under a purchase order to Shape Fidelity, Inc., (80NSSC23PB158) which was awarded by the NSSC for the assembly of the Engineering Development Unit (EDU). The requirement for this contract action is for the assembly of the Qualification Unit.

   The PLA is in support of the SLS Rocket. The PLA must be capable of multiple configurations in a cost effective and schedule friendly manner to accommodate varying mission and payload requirements that drive design changes to the adapter. Hard tooling assembly approaches require design, procurement, fabrication, and assembly of components, significantly increasing lead time between a configuration design release and assembly of the structure. Determinant assembly methods are becoming increasingly common across the aerospace industry to reduce design cycle lag and schedule impacts, reduce tooling costs, and increase flexibility. Determinant assembly is a method that uses the individual components geometry and their relationship to each other to control the final state of the assembly.

The anticipated period of performance for this effort is four months. The estimated value for this procurement is $275k.

4. **FAR 6.303-2(b)(4) – An identification of the statutory authority permitting other than full and open competition:**

The statutory authority permitting other than full and open competition is 10 U.S.C. 3204(a)(1), as implemented by FAR 6.302-1, Only one responsible source and no other supplies or services will satisfy agency requirements.

5. **FAR 6.303-2(b)(5) – A demonstration that the proposed contractor's unique qualifications or the nature of the acquisition requires use of the authority cited:**

The rationale supporting the use of 10 U.S.C. 3204(a)(1) is as described below.

Background: This procurement is for similar services that was awarded by NSSC (Purchase Order 80NSSC23PB158) to Shape Fidelity, Inc., in FY23. The prior procurement was for the award of the Engineering Development Unit (EDU) PLA. This procurement is for the Qualification PLA which will be used for qualification and verification of the PLA hardware.

Shape Fidelity, Inc., successfully performed on the prior contract and utilized their unique specialized engineering techniques and proprietary methods in successful completion of the work.

Rationale: The services procured will entail the use of a series of highly specialized engineering techniques, including but not limited to: (1) structured light scanning, (2) digital assembly, and (3) Computer Aided Design (CAD) guided drilling. These services will be employed in conjunction with one another to achieve determinant assembly of the full-scale SLS Block 1B Payload Adapter article. The engineering approach to be used includes proprietary methods which have been developed and proven by Shape Fidelity, Inc., via a feasibility study on full-scale Payload Adapter hardware. As such, the services to be procured are not available from another provider.

6. **FAR 6.303-2(b)(6) – A description of the efforts made to ensure that offers are solicited from as many potential sources as practicable, including whether a notice was or will be publicized as required by Subpart 5.2 and, if not, which exception under 5.202 applies:**

A notice to the Government Point of Entry (GPE) website (Sam.gov) was published on September 12th, 2024 in accordance with FAR Subpart 5.2. This posting informed potential sources of NASA's intent to award this sole-source contract to Shape Fidelity, Inc. The results of this synopsis are summarized in Section 10 below.

7. **FAR 6.303-2(b)(7) – A determination by the contracting officer that the anticipated cost to the Government will be fair and reasonable:**

The Contracting Officer's signature on this document indicates that the Contracting Officer has determined that the anticipated cost to the government will be fair and reasonable. The Contractor shall be required to submit a proposal to be evaluated and negotiated by the Government. In accordance with FAR 15.404-1(b)(2)(v), fair and reasonable pricing will be based on comparison to the government estimate and historical pricing. Prior to execution of the contractual instrument, a proposal analysis will be performed. The proposal analysis will ensure that the final agreed-to price for the contract action is fair and reasonable.

8. **FAR 6.303-2(b)(8) – Description of the market research conducted, and the results, or a statement of the reasons market research was not conducted (e.g., urgent, and compelling requirement):**

This procurement is procuring similar services that was procured under a previous NASA award (Purchase Order 80NSSC23PB158) with Shape Fidelity Inc., which was awarded by the NSSC for the assembly of the Engineering Development Unit (EDU). This contract action is for the assembly of the Qualification Unit. The prior award was also a sole source award which was synopsized and received no consideration from other offerors.

In performing market research, the CO discussed the requirement with the COR and as it is non-commercial in nature and the unique services required involve Shape Fidelity Inc's proprietary methods there are no other offerors who can meet the requirement.

The above market research demonstrates that awarding to a contractor other than Shape Fidelity Inc., would incur substantial duplication of cost and unacceptable delays to meet agency requirements.

9. **FAR 6.303-2(b)(9) – Any other facts supporting the use of other than full and open competition:**

None.

10. **FAR 6.303-2(b)(10) – A listing of the sources, if any, that expressed an interest in writing in the acquisition:**

A notice of NASA's intent to award this sole-source action was synopsized on the GPE website (Sam.gov) per FAR Subpart 5.2 (See Section 6 above). The presolicitation notice synopsizing NASA's intent to sole source this award was posted on September 12tth, 2024 and closed September 27th, 2024 with no requests for consideration being received.

11. **FAR 6.303-2(b)(11) – A statement of actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the supplies or services required:**

The Agency will continue to examine the market in the future for alternative solutions or new sources before executing any subsequent acquisitions for the same requirements.

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<u>SIGNATURE PAGE</u>

I certify that the facts presented in this justification are accurate and complete.

Kenya Wallace Digitally signed by Kenya Wallace Date: 2024.10.01 14:06:46 -05'00'

EM42 Monique Wallace
Technical Representative

I hereby certify that the above justification is complete and accurate to the best of my knowledge and belief.

Joshua Stiles Digitally signed by Joshua Stiles Date: 2024.10.21 09:49:59 -05'00'

Joshua Stiles
PS22 Contracting Officer